Algebraic Geometry Princeton Quals Questions

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May 04, 2026

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Question 0.0. (Kollar-2001) *Compute the genus of the curve* $x^3 = y^6 + 1$

Question 0.1. (Katz-2000) *Define the genus of a curve in every way that you know how. Why are these equal?*

Question 0.2. (Katz-2000) *State Serre Duality.*

Question 0.3. (Katz-2000) *Define the Hilbert Polynomial. What are the degree, the leading coefficient, and the last coefficient of the Hilbert polynomial?*

Question 0.4. (Katz-2000) *What is the Fermat curve? Compute the genus of the Fermat curve in every way you can. Give a basis of its holomorphic differentials.*

Question 0.5. (Katz-2000) *What does adjunction say in the case of the Fermat curve? How about a degree* d *hypersurface?.*

Question 0.6. (Katz-2000) *What are such hypersurfaces called (*$K3$ *surfaces)?*

Question 0.7. (Katz-2000) *What happens if I know the number of points of the variety over a finite field with* p^r *elements for all* r*?*

Question 0.8. (Katz-2000) *Suppose a variety has exactly* $p^n + 1$ *points over a finite field of characteristic* p^n*. Specifically, consider the projective* n*-space,* P^n*, over a finite field with* p^n *elements.*

What can we deduce about this variety?

Do you know how to prove this?

Question 0.9. (Katz-2000) *State the Weil Conjectures*

Question 0.10. (Katz-2000) *Suppose that my equations are defined over $\mathbb{Z}$. What can I say when I base change to various fields?*

Question 0.11. (Katz-2000) *State Riemann-Roch. What is Riemann Roch good for?*

Question 0.12. (Katz-2000) *What's the relationship between divisors and sections of invertible sheaves?*

Question 0.13. (Katz-2000) *What would Weierstrass say about elliptic curves?*

Question 0.14. (Katz-2000) *Define the Weierstrass p-function. What are its basic properties? What can you say about its poles? Does it satisfy a differential equation?*

How would you know that there should exist such a differential equation?

Question 0.15. (Katz-2000) *Give me a 30 second sketch of everything you know about Riemann-Roch in higher dimensions. What is this useful for?*

Question 0.16. (Bhatt-2025) *Embed a genus 1 curve in projective space.*

Question 0.17. (Venkatesh-2025) *Why is it called an elliptic curve?*

Question 0.18. (Bhatt-2025) *What if it's over $\mathbb{R}$ and has no $\mathbb{R}$-points? Then can you embed in $\mathbb{P}^2$? What about a finite field?*

Question 0.19. (Bhatt-2025) *How can you always find an $\mathbb{F}_p$ point of a genus 1 curve?*

Question 0.20. (Bhatt-2025) *Can I embed a genus 1 curve in higher P^N as a complete intersection?*

Question 0.21. (Bhatt-2025) *Give an example of a smooth affine curve which has nontrivial Picard group.*

Question 0.22. (Bhatt-2025) *Can $\mathrm{Pic}(X)$ be finitely generated? Work over $\mathbb{C}$ and countable field cases.*

Question 0.23. (Bhatt-2025)

Question 0.24. (Bhatt-2025) *If I blow up X surface at a point how does the cohomology change?*

Question 0.25. (Bhatt-2025)

Question 0.26. (Bhatt-2025) *1*

Question 0.27. (Faltings-1992) *What is a scheme?*

Question 0.28. (Faltings-1992) *Define the genus of a curve (in a couple of ways). Explain how to compute it using the Hilbert polynomial. Why is this the same as the dimension of the space of global 1-forms?*

Question 0.29. (Faltings-1992) *or a coherent sheaf F on P^r associated to the graded module M, why is the Hilbert polynomial H(n) (:= Euler characteristic of F(n)) equal to the dimension of M_n for n >> 0?*

Question 0.30. (Faltings-1992) *Compute the genus of the curve y^2 = x^5 - x by using Hurwitz' formula.*

Question 0.31. (Xu, Chengyang-2022) *What do you know about minimal model program on rational surfaces. Why are they different? How do you show they are connected by blow-up and blow-down?*

Question 0.32. (Xu, Chengyang-2022) *What is the class of the canonical divisor on $\mathbb{F}_n$?*

Question 0.33. (Xu, Chengyang-2022) *In general, consider a ruled surface $f : X = \mathbb{P}(E) \to B$ on a curve B of higher genus, E is a rank 2 vector bundle, what is the class of the canonical divisor.*

Question 0.34. (Xu, Chengyang-2022) *What is the class group of a smooth affine conic curve (say, in characteristic 0).*

Question 0.35. (de Jong-1998) *What is an elliptic curve?*

Question 0.36. (de Jong-1998) *Write down a cubic plane curve. Where is the identity? Define the group law for this curve. How do you define it geometrically?*

Question 0.37. (de Jong-1998) *How do you define the group law for a nodal cubic plane curve? How do you do it geometrically?*

Question 0.38. (de Jong-1998) *How would you define a family of curves? Why do you want flatness?*

Question 0.39. (de Jong-1998) *Suppose you have a not-necessarily flat family over a base scheme. How would you impose conditions to make it a flat family?*

Question 0.40. (de Jong-1998) *Take a quadric and a cubic hypersurface in P^3 and intersect them. Suppose the resulting curve is as nice as you want. What is its genus?*

Question 0.41. (de Jong-1998) *State the adjunction formula.*

Question 0.42. (de Jong-1998) *Define genus. What is the canonical sheaf?*

Question 0.43. (Kollar-2015∗) *Compute the genus of* $y^3 = x^6 - 1$

Question 0.44. (Kollar-2015∗) *If X is a projective scheme over the rationals Q, K a finite extension of Q, and X' the base change of X to K, can you relate the cohomology of the structure sheaf of X' to the cohomology of the structure sheaf of X? What if we just look at H^0? How about for H^0 and if we let X = Spec Q[x]/(x^2+1) and K a field containing* $\mathbb{Q}[i]$*?*

Question 0.45. (Faltings-1996∗) *What is an elliptic curve? Compute the genus of* $y^2 = x^5 - x$ *Can a genus 2 curve embed the plane? Prove what you need*

Question 0.46. (Faltings-1996∗) *Give a lower bound for the degree of a map from a genus g curve to* $\mathbb{P}$*^1. Can there be an upper bound?*

Question 0.47. (Faltings-1996∗) *How would you write down all the holomorphic differentials on the Fermat curve?*

Question 0.48. (Faltings-1996∗) *Give an example of a non-hyperelliptic curve. Explain.*

Question 0.49. (Faltings-1996∗) *What is the Jacobian of a curve? How do you determine its dimension?*

Question 0.50. (Faltings-1996∗) *What is the theorem of the cube? What is it used for?*

Question 0.51. (Faltings-1996∗) *Compute the cohomology of projective space.*

Question 0.52. (Faltings-1996∗) *Does every curve embed P^3? Why are all curves even projective?*

What about embedding a curve into P^2?

Question 0.53. (Faltings-1996∗) *What is a regular scheme? Is regularity the same as smoothness over any base field?*

Question 0.54. (Faltings-1996∗) *Do you know what Neron minimal models are? Determinant of cohomology?*

Question 0.55. (Kollar-2001) *Define arithmetic genus. Prove the genus formula. State the explicit computation giving* $H^i(\mathbb{P}^n, \mathcal{O}(m))$ *and Serre duality*

Question 0.56. (Shou-Wu-2021) *Tell me the definition of a scheme, give examples of affine scheme, projective scheme and a scheme that is not affine nor projective.*

Question 0.57. (Shou-Wu-2021) *Why is X=A^2-{0} not projective nor affine? what does the cohomology of affine/projective scheme look like?*

Question 0.58. (Shou-Wu-2021) *So we have a map* f *from X to* $\mathbb{P}^2$*, there is a G_m action on X,* f *is an affine map? The push forward of the structure sheaf of X has a decomposition?*

Question 0.59. (Shou-Wu-2021) *Define curves(smooth) and genus. Classify the curves of genus 0, 1, 2, 3*

Question 0.60. (Tang-2021) *Why is the elliptic curve of degree 3?*

Question 0.61. (Tang-2021) *Give the map from elliptic curve to P^2. How do you find that line bundle?*

Question 0.62. (Boyu-2021) *what is the definition of hyperelliptic curves*

Question 0.63. (Boyu-2021) *So is there a meromorphic function from a curve of lower genus to higher genus?*

Question 0.64. (Shou-Wu-2021) *compute the genus of the curve* $y^5 = x(x-1)(x-\lambda)$

Question 0.65. (Katz-2003∗) *State Riemann-Roch for curves and Riemann Surfaces*

Question 0.66. (Katz-2003∗) *Define the arithmetic genus and geometric genus of a variety.*

Question 0.67. (Katz-2003∗) *What can you say about global sections of* $\Gamma(X, \omega)$*? Sections of* $H^1(X, O)$*?*

Question 0.68. (Katz-2003∗) *What is the genus of $y^2 = x^{11} - 1$? How about $y^2 = f(x)$ with $f(x)$ of odd degree and distinct roots? Give the general genus formula for that curve*

Question 0.69. (Katz-2003∗) *Talk about intersection theory for a surface.*

Question 0.70. (Katz-2003∗) *Why does an elliptic curve have a cubic form in P^2?*

Question 0.71. (Katz-2003∗) *State Abel's Theorem for an elliptic curve. What about higher dimensional varieties?*

Question 0.72. (Katz-2003∗) *What does Hasse's Theorem say about the number of points on an elliptic curve over a finite field.*

Question 0.73. (Katz-2003∗) *How does this relate to the Weil Zeta Function of a projective variety? What does this zeta function look like for an elliptic curve? What does it look like for P^n?*

*(**Hint:** Identical Denominator)*

Question 0.74. (Katz-2003∗) *Where does the coefficient a_p come from in the elliptic curve's Zeta function?*

Question 0.75. (Katz-2003∗) *Suppose that you have a large finite set of equations over a finite field and the number of points on the resulting variety over $\mathbb{F}_{q^n}$ for all n. Further suppose that you know that the resulting variety is a hypersurface. How might one determine the genus of this hypersurface?*

(Hint: Look at the numerator of some zeta function.)

Question 0.76. (Katz-2003∗) *Why does an elliptic curve have a cubic form in P^2?*

Question 0.77. (Ellenberg-2005) *Describe the Jacobian of a curve.*

Question 0.78. (Ellenberg-2005) *What's a sheaf? What's a morphism of schemes? What's the push-forward? pull-back? What's a quasi-coherent sheaf? coherent sheaf?*

Question 0.79. (Ellenberg-2005) *Suppose F is coherent on Y when is the push-forward of F coherent on X?*

Question 0.80. (Ellenberg-2005) *What's the geometric reason a map is called finite? Give a map that is quasi-finite but not finite.*

Question 0.81. (Ellenberg-2005) *Can you show that all plane conics are birational to a line?*

Question 0.82. (Ellenberg-2005) *If I draw N points in the plane, can you tell me if there's a conic that goes through them?*

Question 0.83. (Ellenberg-2005) *Tell me about the analogy of the class group for function fields over F_p?*

Question 0.84. (Pandharipande-2003) *What do you know about vector bundles on* $\mathbb{P}^1$

Question 0.85. (Pandharipande-2003) *What are all line bundles on P^2? (i had to prove that any line bundle on P^2 is* $sO(n)$ *for some* n.

Question 0.86. (Pandharipande-2003) *Is the tangent bundle of P^2 a direct sum of line bundles?*

Question 0.87. (Pandharipande-2003) *Do you know what are chern classes? calculate the chern classes of tangent bundle of P^2.*

Question 0.88. (Pandharipande-2003) *Is it possible to have a family, parametrized by C, of rank 2 vector bundles on P^1 which is trivial at every point of C except the origin?*

Question 0.89. (Pandharipande-2003) *What do you know of global sections of these vector bundles?*

Question 0.90. (Katz-1997∗) *Define an algebraic group. Give some examples. Which of them are rational varieties? Prove it.*

Question 0.91. (Katz-1997∗) *Define the algebraic group O(n) as xx^t=1. Give a more intrinsic definition of this, via nondegenerate symmetric bilinear forms. Give intrinsic definitions of Sp, U, etc. in this way. Which ones are connected?*

Question 0.92. (Shou-Wu-2025) *What is an algebraic curve? Give the modern smooth projective definition.*

Question 0.93. (Shou-Wu-2025) *What are genus zero curves?*

Question 0.94. (Shou-Wu-2025) *What is* $\mathrm{Aut}(\mathbb{P}^1)$*?*

Question 0.95. (Shou-Wu-2025) *What are the line bundles on elliptic curves? Why are their group laws algebraic? What breaks in genus 2?*

Question 0.96. (Venkatesh-2025) *Pick your favorite curve, not genus 1 or 0 and compute its genus.*

Question 0.97. (Kollar-2017) *Define an affine scheme. How do you tell if a scheme is affine? How else can you show that a scheme isn't affine? Why isn't P^n affine? There are lots of ways, give some.*

Question 0.98. (Kollar-2017) *Why doesn't P^n have any global sections other than k?*

Question 0.99. (Kollar-2017) *What nice properties does P^n have? If you take the first projection X x P^n –> X, what can you say about it?*

Question 0.100. (Kollar-2017) *What can you say about the image of a closed subset?*

Question 0.101. (Kollar-2017) *Can you prove that the structure morphism* $P^n \longrightarrow \text{Spec } k$ *is proper?*

Question 0.102. (Kollar-2017) *What can you say about the curve* $y^3 = x^6 + 1$*? Assume the characteristic is not 2 or 3. Compute the genus. What would a topologist see in this, and how do you see this directly from the equation?*

Question 0.103. (Kollar-2006) *Talk about lines contained in hypersurfaces of degree 3 in the projective space. Why is this a variety?*

Question 0.104. (de Jong-1996) *Define an affine scheme. Define a variety. What additional conditions would some people impose?*

Question 0.105. (de Jong-1996) *Can you give an irreducible variety that is not absolutely irreducible?*

Question 0.106. (de Jong-1996) *Define the genus of a (projective) curve.*

Question 0.107. (de Jong-1996) *Give another definition of genus.*

Question 0.108. (de Jong-1996) *Define the Hilbert polynomial.*

Question 0.109. (Katz-1996) *What is a differential? What can you say about this sheaf?*

Question 0.110. (Katz-1996) *Can you state Serre duality? Can you state it for curves? What can you say about the residues of a 1-form on a curve?*

Question 0.111. (Katz-1996) *How would you compute H^1 of a curve? What's the fewest open sets you can use? Describe Serre duality in terms of this cover.*

Question 0.112. (Katz-1996) *Do you know what the Cayley transform is?*

Question 0.113. (Katz-1996) *How can you show that a curve is not rational? In fact, if you have a rational map between two curves, what can you say? Now what about higher-dimensional varieties? Can a rational variety map to a curve of positive genus?*

Question 0.114. (Katz-1996) *Define an elliptic curve. What is Weierstrass form? What is special about it? What is a flex? How many flexes are there? Can a line touch an elliptic curve to order higher than 3? How would I tell whether a given point has a given finite order. Prove that the curve (with the geometric group law) is isomorphic to Pic^0.*

Question 0.115. (Pandharipande-2002) *Can P^1 map to a curve of higher genus?*

Question 0.116. (Pandharipande-2002) *What about curve of genus g1 map to curve of g2 ? When can you have g1=g2?*

Question 0.117. (Pandharipande-2002) *Compute the cohomology for the normal bundle of the rational normal curve in P^3.*

Question 0.118. (Pandharipande-2002) *Let's talk about a curve which is a complete intersection of $\mathbb{P}^n$. Given the degrees, what's the genus?*

Question 0.119. (Pandharipande-2002) *Let's think about surfaces. How many types of quadric surfaces in P^3 do you have?*

Question 0.120. (Pandharipande-2002) *Is the quadric cone rational? Y. What about a cubic hypersurface in P^3.*

Question 0.121. (Pandharipande-2002) *For a curve of genus g, what's the smallest d such that any line bundle of degree g >= d has some global section?*

Question 0.122. (Kollar-2008) *Calculate the genus of plane curve defined by y^3=x^6-1.*

Question 0.123. (Kollar-2014) *Suppose X is a scheme over a field K, L is a field extension of K, then what can you say about cohomology of coherent sheaves on X and on Xtimes_K L?*

Question 0.124. (Kollar-2014) *State the semi-continuity theorem. Do you know an example that the dimension of cohomology groups jumps?*

Question 0.125. (Kollar-2014) *Consider an elliptic curve, what is the cohomology for a divisor?*

Question 0.126. (Kollar-2014) *Define a Fano variety. What is your favorite Fano variety? Is the blow up of a point in P^3 Fano?*

Question 0.127. (Kollar-2014) *Let's look at the intersection of anti-canonical divisor with curves. What about P^3 blowing up two points?*

Question 0.128. (Kollar-2014) *Do you know some properties of rational curves in Fano varieties? Can you prove some result?*

Question 0.129. (Kollar-2019) *Study smooth curves of genus 0 (over a field that is not necessarily algebraically closed).*

Question 0.130. (Sawin-???) *Write down a genus 25 curve. (Hyperelliptic* $y^2 = f(x)$ *where f is degree 51 or 52)*

Question 0.131. (Bhargava-???) *What happens if you treat hyperelliptic* $y^2 = f(x)$ *where f is degree 51 or 52 as a plane curve? How would you resolve it?*

Question 0.132. (Bhargava-???) *Is every genus g curve a plane curve? What if g is already a triangle number? Take a small g, What's the dimension of* $M_{\{g,n\}}$*?*

Question 0.133. (Bhargava-???) *Take g=3. Why are all genus 3 curves not plane curves?*

Question 0.134. (Sawin-???) *What does the degree-genus formula compute if it's not smooth?*

Question 0.135. (Sawin-???) *Say you have a curve C and a double curve 2C. You would expect the Euler characteristic of 2C to be double that of C, but this is not the case. Why?*

Question 0.136. (Sawin-???) *Compute the cohomology of the cotangent sheaf of P^n. Why is the canonical bundle O(-n-1)?*

We stopped at MICHAEL MCBREEN ALGEBRAIC GEOMETRY QUALS

Question 0.137. (Kollar-2019)

Question 0.138. (Kollar-2019)

Question 0.139. (Kollar-2019)

Question 0.140. (Kollar-2019)

1 Notes

Careful with Shonwu, pretty recent

A lot of Harthstone references in Katz

A lot of Hurwtiz formula

A lot of Harthsone and Rising Sea